

Laboratory 6: Conditional Image Generation

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1 Introduction

We present this report on the work carried out in Laboratory 6 on conditional image generation. Two different approaches are studied.

In the first task, we cover *artistic style transfer* using the CNN-based algorithm proposed by Gatys *et al.* [1]. Given a content image and a style image, the method synthesises a new image that preserves the semantic structure of the former while adopting the visual texture and colour palette of the latter. The key insight is that a deep convolutional network trained for object recognition implicitly separates content, encoded in high-level feature activations, from style, encoded in the second-order statistics of those activations across multiple layers.

Moreover, in the second task, we cover *facial animation* using GANimation [2], a GAN-based method that animates a single RGB portrait by conditioning the generator on Action Unit (AU) annotations. AUs describe facial muscle movements continuously, allowing fine control over the intensity and combination of different expressions without requiring paired training data.

Both tasks are analysed theoretically, with the loss functions identified in the corresponding code. For the artistic style task, experiments are also run and the results are evaluated qualitatively.

2 Methods

2.1 Artistic Style

2.1.1 Overview of the Algorithm

The neural style-transfer algorithm of Gatys *et al.* [1] solves an *image-optimisation* problem: starting from a white-noise image $\vec{x}$, gradient descent is applied to minimise a composite loss that simultaneously penalises deviation from the *content* of a reference photograph $\vec{p}$ and from the *style* of a reference artwork $\vec{a}$. The network weights are **kept fixed throughout**; only the pixel values of $\vec{x}$ are updated.

2.1.2 Content Representation and Content Loss

Each convolutional layer l with N_l filters produces N_l feature maps of spatial size M_l (height $\times$ width). The responses are stored in a matrix $F^l \in \mathbb{R}^{N_l \times M_l}$, where F_{ij}^l is the activation of filter i at spatial position j .

Higher layers encode *what* is in the image (objects, their arrangement) while discarding exact pixel detail; lower layers preserve fine-grained pixel information. The paper uses layer **conv4.2** as the content layer because it retains high-level structure without sacrificing all spatial detail.

The **content loss** measures the mean-squared difference between the feature responses of the generated image $\vec{x}$ and the content image $\vec{p}$ at layer l :

$$\mathcal{L}_{\text{content}}(\vec{p}, \vec{x}, l) = \frac{1}{2} \sum_{i,j} (F_{ij}^l - P_{ij}^l)^2, \quad (1)$$

where P^l and F^l are the feature representations of $\vec{p}$ and $\vec{x}$ at layer l , respectively. Minimising this loss drives the generated image to produce similar activations, thereby preserving the semantic content.

2.1.3 Style Representation and Style Loss

Style is captured not by *where* features activate but by *how they co-activate* across the spatial extent of a layer. This is formalised through the **Gram matrix**:

$$G_{ij}^l = \sum_k F_{ik}^l F_{jk}^l, \quad (2)$$

which contains the inner product between the vectorised feature maps i and j in layer l . The Gram matrix is therefore a $N_l \times N_l$ correlation matrix that captures second-order statistics of filter responses, discarding spatial arrangement and acting as a texture descriptor.

The style loss for a single layer compares the Gram matrix of the generated image G^l with that of the style image A^l :

$$E_l = \frac{1}{4N_l^2 M_l^2} \sum_{i,j} (G_{ij}^l - A_{ij}^l)^2. \quad (3)$$

The total style loss aggregates contributions from a set of layers $\mathcal{L}$ weighted by w_l :

$$\mathcal{L}_{\text{style}}(\vec{a}, \vec{x}) = \sum_{l \in \mathcal{L}} w_l E_l. \quad (4)$$

In practice, layers `conv1_1`, `conv2_1`, `conv3_1`, `conv4_1`, and `conv5_1` are used with equal weights $w_l = 1/5$. Including multiple scales allows the style representation to capture textures at different granularities.

2.1.4 Total Loss and Optimisation

The two objectives are combined into a single **total loss**:

$$\mathcal{L}_{\text{total}}(\vec{p}, \vec{a}, \vec{x}) = \alpha \mathcal{L}_{\text{content}}(\vec{p}, \vec{x}) + \beta \mathcal{L}_{\text{style}}(\vec{a}, \vec{x}), \quad (5)$$

where α and β are scalar weights that control the content-style trade-off. A large ratio α/β preserves content at the expense of style fidelity; a small ratio emphasises stylisation. Typical values used in [1] are $\alpha/\beta \in \{10^{-3}, 10^{-4}\}$.

Gradients of $\mathcal{L}_{\text{total}}$ with respect to the pixel values of $\vec{x}$ are computed via standard back-propagation and used to update $\vec{x}$ iteratively (L-BFGS or Adam optimiser). The network weights are never modified.

2.1.5 Identification of the Loss in the Code

The total loss defined in Equation (5) is implemented in `main.m`. The content loss (Eq. 1) appears in the following lines:

```
function loss = contentLoss(transferContentFeatures,contentFeatures,contentWeights)
    loss = 0;
    for i=1:numel(contentFeatures)
        temp = 0.5 .* mean((transferContentFeatures{1,i} - contentFeatures{1,i}).^2,"all");
        loss = loss + (contentWeights(i)*temp);
    end
end
```

The Gram matrix (Eq. 2) is computed as:

```
function gramMatrix = calculateGramMatrix(featureMap)
    [H,W,C] = size(featureMap);
    reshapedFeatures = reshape(featureMap,H*W,C);
    gramMatrix = reshapedFeatures' * reshapedFeatures;
end
```

The style loss per layer (Eq. 3) is:

```

function loss = styleLoss(transferStyleFeatures,styleFeatures,styleWeights)
    loss = 0;

    for i=1:numel(styleFeatures)
        tsf = transferStyleFeatures{1,i};
        sf = styleFeatures{1,i};
        [h,w,c] = size(sf);

        gramStyle = calculateGramMatrix(sf);
        gramTransfer = calculateGramMatrix(tsf);
        sLoss = mean((gramTransfer - gramStyle).^2,"all") / ((h*w*c)^2);
        loss = loss + (styleWeights(i)*sLoss);
    end
end

```

Finally, the weighted combination (Eq. 5) is:

```
loss = (params.alpha * cLoss) + (params.beta * sLoss);
```

2.2 Facial Animation

2.2.1 Overview of GANimation Architecture

The second task utilizes GANimation [2] to animate a single facial photograph using continuous muscle activations called Action Units (AUs). The architecture isolates tasks into two primary neural blocks:

- **The Generator (G):** Receives the source image and a target vector $\vec{y}$ of desired Action Units. Rather than generating a full scene from scratch, it yields a *Color Content Map* (C) and an *Attention Mask* (A). The final composition is blended as:

$$\vec{I}_{\text{final}} = A \cdot C + (1 - A) \cdot \vec{I}_{\text{original}} \quad (6)$$

- **The Discriminator (D):** Evaluates outputs using two heads: an *Adversarial Critic* (D_{adv}) to verify realistic skin textures, and an *Expression Regressor* (D_{reg}) to estimate Action Unit values and check expression accuracy.

2.2.2 Identification of the Cost Functions in the Code

Following the implementation code repository, the training routine optimizes a joint objective function. The total costs for the Generator and Discriminator handle four distinct loss terms:

1. **Adversarial Loss** (criterion_{GAN}): *Measures image realism using an LSGAN (Mean Squared Error) or WGAN*

```
loss_D_g_gan = criterion_GAN(D_g_gan, True)
```

Expression Loss (criterion_{wreg}): *Evaluates structural regression by computing the meansquared distance between input and output facial images*

```
loss_G_g_reg = criterion_wreg(D_g_reg, target_au)
```

Identity / Cycle Consistency Loss (criterion_{id}): *Guarantees identity tracking by passing the generated facial canvas level structural L1 distances:*

```
loss_G_back = criterion_id(real_img_back, real_img)
```

Attention Mask Loss: Enforces an L1 normalization penalty directly on the attention mask array pixels to minimize distortion and keep moving zones tightly localized:

```
loss_G_mask = torch.mean(torch.abs(attention_mask))
```

2.2.3 Failure Cases and Technical Limitations

Analyzing the network reveals two distinct scenario limitations:

- **Extreme Head Poses:** When subjects turn completely to the side, features distort because the flat 2D architecture lacks 3D geometric depth knowledge, causing the expression regressor to lose tracking markers.
- **Partial Face Occlusions:** When objects like hands or thick glasses cover skin tissue, the generator treats them as face parts and attempts to warp them into expressions, causing unnatural visual artifacts.

2.2.4 Theoretical Upgrades

To resolve these errors, two architectural improvements can be proposed from a theoretical standpoint:

1. **3D Morphable Models (3DMM):** Fitting a 3D physical mesh structure over the input face would allow Action Units to deform along a 3D topology, keeping expressions stable across extreme profile angles.
2. **Semantic Segmentation Masking:** Incorporating a pre-processing segmentation layer to identify non-face objects (such as hands or glasses) would allow the attention module to lock those pixel zones completely unchanged.

3 Experiments

3.1 Artistic Style

3.1.1 Experimental Setup

All experiments used the VGG-19 network with average-pooling. The content image was `lighthouse.png`. Two style images were selected from the provided data:

- **Style 1:** `starryNight.jpg` - *The Starry Night* by Vincent van Gogh (1889).
- **Style 2:** `dali.jpg` - *The Persistence of Memory* by Salvador Dalí (1931).

Experiments were run for **100 iterations** on CPU. The content layer was `conv4_2` and the style layers were `conv1_1` through `conv5_1`. The ratio $\alpha/\beta = 10^{-3}$ was used for both experiments.

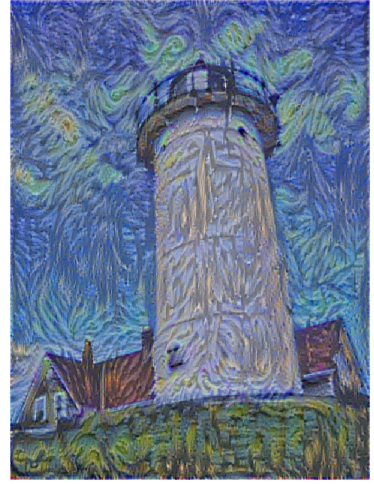


Figure 1: Content image: `lighthouse.png`.

3.1.2 Results



(a) Style image: *The Starry Night*.



(b) Generated image (100 iter.).

Figure 2



(a) Style image: *The Persistence of Memory*.



(b) Generated image (100 iter.).

Figure 3

3.1.3 Qualitative Evaluation

Experiment 1: *The Starry Night* style. From the generated image, the lighthouse structure is clearly recognisable. For instance, the white tower, the red-roofed buildings, and the grassy foreground are all preserved. Moreover, the most visible effect of the style transfer is in the texture and colour palette. The sky and the surface of the tower have adopted the swirling, wave-like brush strokes characteristic of van Gogh’s painting, and the dominant colours have shifted towards the blues and yellows of *The Starry Night*. The semantic content is therefore well maintained, while the artistic texture is clearly transferred. At 100 iterations the result still shows some roughness, especially at the edges of the tower. Running for more iterations would likely produce a smoother output with stronger stylisation, but the current result already demonstrates the algorithm’s ability to blend content and style effectively.

Experiment 2: *The Persistence of Memory* style. Again, the lighthouse structure is preserved, but the colour palette has changed significantly. The warm sandy browns and orange tones of Dalí’s painting dominate the output, replacing the original blue sky and green grass. The textures are less sharp than in Experiment 1, which reflects the flatter, more blended surface style of Dalí’s painting compared to

the thick brush strokes of van Gogh. Some fine details of the lighthouse, such as the windows, are harder to distinguish in this result, suggesting that the Dalí style is more aggressive in overwriting low-level detail.

Comparison between the two styles. Both experiments confirm that the algorithm successfully separates and recombines content and style. The van Gogh transfer produces a more visually striking result because its swirling textures create a strong contrast with the original photograph. The Dalí transfer is subtler: the content is equally preserved, but the output looks closer to a colour-graded photograph than to a painting. This difference is explained by the style images themselves: van Gogh’s work has highly structured, repeated brush-stroke patterns that the Gram matrices capture easily, while Dalí’s painting has smoother gradients and less repetitive texture.

4 Conclusion

In this laboratory, we have explored two complementary strategies for conditional image generation based on deep learning.

For the artistic style transfer task, the algorithm of Gatys *et al.* [1] was studied and implemented. The method successfully disentangles content and style by using the feature activations of a fixed VGG-19 network: content is preserved through layer `conv4_2`, while style is captured by Gram matrices computed across five convolutional layers. The total loss in Equation (5) balances both objectives and is minimised by updating only the pixel values of the generated image. The experiments showed that both styles were transferred convincingly at 100 iterations on CPU. The van Gogh style produced a visually striking result with clear swirling textures and a blue-yellow palette, while the Dalí style produced a subtler output dominated by warm tones and smooth gradients. In both cases the content of the lighthouse image remained recognisable, confirming that the chosen ratio $\alpha/\beta = 10^{-3}$ correctly prioritises content preservation. More iterations or GPU computation would further improve the quality of the generated images.

For the facial animation task, the GANimation architecture was analysed theoretically. The generator produces an attention mask and a colour map that are blended to modify only the relevant facial regions, while the discriminator jointly ensures visual realism and expression accuracy through adversarial and regression losses. The main failure cases identified, extreme head poses and face occlusions, arise from the lack of 3D geometric knowledge in the model. Two improvements were proposed to address these limitations: incorporating 3D morphable models to handle out-of-plane rotations, and adding a semantic segmentation step to prevent non-face regions from being incorrectly warped.

References

- [1] L. A. Gatys, A. S. Ecker, and M. Bethge, “A neural algorithm of artistic style,” *arXiv preprint arXiv:1508.06576*, 2015.
- [2] A. Pumarola, A. Agudo, A. M. Martinez, A. Sanfeliu, and F. Moreno-Noguer, “Ganimation: One-shot anatomically consistent facial animation,” *International Journal of Computer Vision*, vol. 128, no. 3, pp. 698–713, 2020.